Novelty decision memo - ClaimAudit / ScopeGuard

19 October 2026. Decision made after Stage 1 and before new implementation or results. **No candidate survives as a defensible patent-novelty claim on the evidence available. Proceed on a research-paper-only route.** Source IDs resolve to the opened-source register in STAGE1_PRIOR_ART.md. This is a screening conclusion, not a legal finding that no patent could ever exist.

Rank 1 - Compile once, condition many times. Feature: compile the complete four-status source-binding relation once into a decision diagram; answer all proposed header resolutions through restrictions of that shared object. Closest art: Bryant's BDD operations (P04), incremental sensitivity analysis (P06), query-guided resolution (P07) and ProvSQL compilation (P16). These already provide the mechanism; no established technical distinction survives. Testable effect: lower total clarification time and peak memory than rebuilding a verifier for each header, at identical decisions, including compilation cost. Falsification: generic BDD restriction implements the feature unchanged (novelty fails); no measured improvement on reserved inputs or adverse growth defeats any general efficiency claim. **Rejected as novelty; selected only as a transparent engineering/research extension.**

Rank 2 - Correlated scientific-scope robustness. Feature: reject a compound claim unless a joint experimental context supports it under every admissible correlated header interpretation. Closest art: BID/and-xor models and certain answers (P01-P03), compound scientific constraints (P13), Claim Cards (P12) and relational-summary verification (P18). A scientific naming convention and a four-status policy do not establish a new algorithm. Testable effect: fewer false supports on independently annotated table claims at matched coverage, relative to NLI and strong LLM verification. Falsification: an ordinary certain-answer encoding yields identical decisions (novelty fails); a matched LLM makes no target errors or extraction errors erase the advantage (utility hypothesis fails). **Rejected as novelty; useful evaluation question.**

Rank 3 - Error-aware evidence rechecking. Feature: prioritize source fragments whose correction changes the downstream claim decision under a verification budget. Closest art: budgeted cleaning (P08), query-guided resolution (P07), worst-case verification-error mitigation (P17) and scientific visual tool selection (P22). Those sources already cover the core mechanism; no verified distinction survives. Testable effect: fewer verifier calls or tokens at a fixed false-support rate. Falsification: existing provenance/error-mitigation policies express the same rule, or measured cost is no lower. **Rejected; not implemented as an alleged invention.**

Next stages. Implement Rank 1 as explicitly known-method engineering, compare with a real BDD possible-world engine, Z3 SMT and independent exhaustive verification, and retain adverse cases. Freeze a new evaluation set before running it; use paper-derived fixtures if verifiable. Report missing independent annotation and LLM access honestly. Re-fill all ten IDF-B questions as a research disclosure with no affirmative novelty claim. Each measured effect is evidence of implementation behavior, not patentability.

CRI and release risks. Official CRI 2025 guidance (L01, sections 3.5/4.5) requires consideration of technical contribution; novelty and inventive step are separate. No measured effect yet exists at this decision checkpoint. Patent-database coverage is incomplete. Obtain institutional IP advice on disclosure timing before public code, manuscript or demo release; no public release is performed here.